

DOCUMENT ROTATION & SKEW ANALYSIS

Optical Character Recognition (OCR) Pre-processing Diagnostic

CONFIDENTIAL • TEST RUN #884

This document evaluates the impact of document orientation skew and rotation angles (1°, 3°, 10°, and 90°) on OCR layout restoration pipelines. Text alignment vectors must follow physical transformation boundaries for accurate bounded-box detection.

1. TRANSFORMATION & ALIGNMENT MATRIX

MINOR FEEDER SKEW ANGLE: 1.0° PATIENT FORM #0912 Status: Slight baseline tilt. Text follows +1° inclination vector. Line bounding box maintained.	STANDARD SCAN OFFSET ANGLE: 3.0° INVOICE #INV-4029 Status: Moderate skew offset. Requires Radon deskew filtering. Baseline displacement noticeable.
SEVERE ANGLE DISTORTION ANGLE: 10.0° LAB REPORT #8821 Status: Heavy rotation. OCR line segmentation fails. Polygon alignment mandatory.	ORTHOGONAL ORIENTATION ANGLE: 90.0° SIDEBAR DATA Status: Landscape shift. Needs 90° CCW correction.

2. OCR ACCURACY IMPACT PERFORMANCE TABLE

Rotation Angle	Hough Line Vector	Character Error Rate (CER)	Bounding Box Fidelity	Auto-Deskew Status
1.0°	[0.0175, 0.9998]	0.42%	99.2%	PASS (Negligible)
3.0°	[0.0523, 0.9986]	2.18%	94.5%	PASS (Corrected)
10.0°	[0.1736, 0.9848]	18.45%	61.0%	WARN (Requires Rescale)
90.0°	[1.0000, 0.0000]	98.90%	0.0%	FAIL (Orientation Flip)

Diagnostic Recommendation: Angles $\leq 3^\circ$ can be normalized via affine transform matrices without quality loss. Angles at 10° or 90° require primary orientation detection (EXIF/Tesseract OSD) prior to segmentation.